

COVID-19 Global Analysis

An Executive Briefing

Period Analyzed: January 1, 2020 - April 30, 2021

Coverage: 219 countries across 6 continents

Data Source: WHO COVID-19 Dashboard via Our World in Data

Prepared for executive decision-makers

1. The Big Picture

By the end of April 2021, the world had recorded over 150 million confirmed COVID-19 cases and more than 3.2 million deaths. But behind that single number is a story of stark inequality - in who got sick, who survived, and who got vaccinated.

This report breaks down the pandemic into actionable themes for decision-makers.

2. The Pandemic Hit Continents Very Differently

Continent	Total Cases	Deaths	Death Rate	Cases/M	Deaths/M
N. America	32.3M	576K	1.78%	97,724	1,741
S. America	14.7M	404K	2.75%	68,964	1,900
Asia	19.2M	212K	1.11%	13,315	147
Europe	5.7M	128K	2.25%	38,907	876
Africa	1.6M	54K	3.44%	7,671	264
Oceania	30K	910	3.05%	1,169	36

Key insight: Africa had the highest death-to-case ratio (3.44%) despite the fewest cases per capita. This signals limited testing capacity masking true infection levels, combined with weaker healthcare infrastructure.

Key insight: South America had the highest per-capita death toll (1,900 per million) - suggesting healthcare system overwhelm and delayed public health responses.

3. The Deadliest Countries

By total death toll

The US, Brazil, India, Mexico, and the UK accounted for more than half of all global deaths.

By case fatality rate (>100k cases)

Country	Death Rate	Key Signal
Mexico	9.25%	Extreme outlier - 4x the global average
Egypt	5.86%	Limited healthcare capacity
Ecuador	4.88%	Early healthcare collapse
China	4.73%	Testing focused on symptomatic cases
Bosnia	4.31%	Older population, weaker system

The Mexico anomaly demands attention: At 9.25%, Mexico's case fatality rate was nearly 5 times higher than India's (1.11%) and 3 times higher than Brazil's (2.75%). Low testing rates,

comorbidities, and hospital access issues combined into a deadly outcome.

4. The Vaccination Race - A Story of Two Worlds

By April 2021, vaccination campaigns were well underway in wealthy nations but barely started elsewhere.

Leaders in full vaccination

Country	Fully Vaccinated	Share of Population
Israel	5.1M	58.84% - global leader
USA	101.4M	30.64%
UK	~15M est.	~22%
Serbia	1.5M	21.87%
Hungary	1.9M	20.17%

Laggards (among large nations)

Country	Fully Vaccinated	Share of Population
India	26.6M	1.93% - massive gap
Indonesia	7.6M	2.80%
Canada	1.1M	2.89%
Russia	7.5M	5.16%

Key insight: Vaccination access correlated strongly with GDP. Wealthier nations secured early vaccine contracts, while developing nations waited. This gap directly predicted the Delta variant wave that devastated India just weeks after this dataset ends.

5. The Quality of Data Problem

The analysis surfaced significant data quality issues that decision-makers must account for:

Issue	Impact
90% missing ICU data	Cannot assess true healthcare system strain
~89% missing vaccination data	Early vaccination picture is incomplete
50% missing testing data	Case counts understated in poor nations
4,111 rows with null continent	Aggregate entries mixed with country data

Bottom line: Countries with better data infrastructure (Europe, North America) appear to have higher case counts partly because they tested more - not necessarily because they had more infections.

6. Transmission Dynamics

The reproduction rate (R) measures how fast the virus spreads. By late April 2021:

1. Highest transmission risk: Nepal (R=1.99), Sri Lanka (1.79), Cambodia (1.57) - second waves building in Asia
2. Lowest transmission: Many island nations (R~0) - successful containment or data gaps
3. Most large economies: R was near or below 1.0 due to vaccination + prior infection immunity

Warning signal: The Asian countries with $R > 1.5$ in April 2021 became the epicenters of the Delta wave in May-June 2021.

7. What This Means for Decision-Makers

Strategic takeaways

1. Testing capacity is strategic intelligence. Countries with low testing (Africa, parts of Asia) operated blind. Investment in surveillance testing is an early-warning system, not a cost.
2. Vaccine equity is not charity - it is epidemiology. As long as large populations remain unvaccinated (India at 1.93%), new variants will emerge that threaten even highly vaccinated nations.
3. Death rates are a lagging indicator of system failure. Mexico's 9.25% CFR and South America's 1,900 deaths/million were predictable months in advance from ICU capacity data - if that data had been collected.
4. Data infrastructure determines response quality. The countries that invested in real-time health data systems made faster, more targeted interventions.

Recommended actions

- Strengthen ICU and hospitalization reporting - this is the metric that determines lockdown timing
 - Track death rates alongside case rates - a diverging CFR signals either a testing gap or a healthcare crisis
 - Monitor R-values regionally, not nationally - subnational R data catches outbreaks before they become waves
 - Prioritize vaccine delivery to high-R countries first - stopping transmission protects everyone
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8. A Note on Data & Methodology

Data: WHO COVID-19 Dashboard via Our World in Data, downloaded April 19, 2026.

Period: January 1, 2020 to April 30, 2021.

Coverage: 85,171 observations across 219 locations, 59 variables.

Processing: ETL via SQL, analysis in Python.

Visualization: Power BI file.

Limitations:

- Case fatality rates are sensitive to testing rates - low-testing countries appear to have higher CFRs
- July 2020 data for some countries contains known recording anomalies
- Vaccination data in early 2021 is sparse (vaccines were only beginning mass rollout)

Report prepared for executive decision-makers.